in first order logic

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour
Total Marks:30

QUESTIONS: 3

Annexure A

Dry Run Evaluation

Code of Conduct:

I POOJASREE B (192411091) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Poojasree B

20+20+24+17+12

93

ARTIFICIAL INTELLIGENCE

POOJASREE B
192411091
CSA1710
CO2_AT3

1. Evaluation Function

An AI game playing agent uses $E(s) = 2M + 3K + 5Q + P$. For S1: $M=4, K=2, Q=1, P=5$. For S2: $M=3, K=5, Q=2, P=4$. Calculate both evaluation scores and determine which state MAX should select.

Given:

State S1

$M=4$

$K=2$

$Q=1$

$P=5$

State S2

$M=3$

$K=5$

$Q=2$

$P=4$

MAX decision

Since MAX selects the state with the higher evaluation score.

Therefore, MAX should select S2.

ANSWER: $S1 = 24$, $S2 = 35$

$\downarrow$

Select S2.

2. Search with Partial Information

A robot must travel from A to G. Initially it knows only $A \to B$ and $A \to C$. On reaching B it discovers $B \to D$ and later $D \to G$. Show how the search space changes as new information is obtained.

Given

Start = A

Goal = G

Initially known : $A \to B$ and $A \to C$

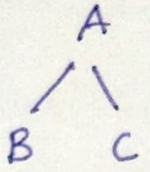
After reaching B, discovers $B \to D$

Later discovers $D \to G$

The search space changes as the robot obtains new information.

Initial search space

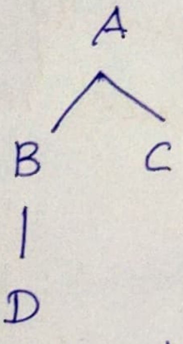
Initially, the robot knows only:



So the robot can consider B and C as possible next states

After reaching B

The robot discovers a new connection $B \rightarrow D$. The search space becomes:



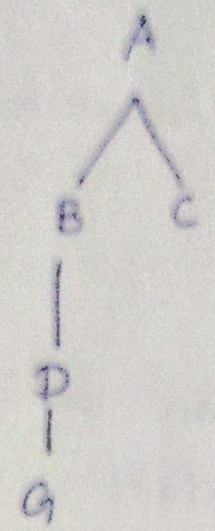
Now D becomes a new reachable state.

After reaching D

The robot discovers:

$D \rightarrow G$

The search space becomes:



The goal G is now reachable

Final path

In partial-information search, the search space is incrementally expanded as the agent discovers new information.

3. Bidirectional Search

Consider the linear graph A → B → C → D → E → F. Start = A and Goal = F. Perform Bidirectional Search, show forward and backward expansion and identify the meeting point.

Given the linear graph:

A → B → C → D → E → F

Start = A Goal = F

Bidirectional search performs two searches simultaneously:

Forward search: Starting from A

Backward search: Starting from F

Step 1 - Initial expansion

Forward: A

Backward: F

Step 2 - Expand both sides

Forward: A → B

Backward: F → E

Step 3 - Expand

Forward: A → B → C

Backward: F → E → D

Step 4 - Meeting point

The forward search continues:

A → B → C → D

The backward search have already reached D:

F → E → D

Therefore, both searches meet at: ..

Search diagram

Forward search

Backward search

$$A \rightarrow B \rightarrow C \rightarrow D \leftarrow E \leftarrow F$$

$\downarrow$

Meeting point

Final path

combining both searches:

Meeting point = D